

Measures of Inequality

Diploma Thesis

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Brno 2024

Acknowledgement:

I would like to first thank my supervisor Prof. Marek Litzman, Ph.D. for his support and helpful feedback throughout our discussions. His guidance has been a great help in writing this thesis and it would not have been completed without his expertise.

I am also grateful to the staff of the Faculty of Business and Economics, Mendel University in Brno. Their hard work and dedication have provided me with valuable knowledge and skills to finalize this paper. Special thanks to my professors and classmates whom I met during my studies for their support and encouragement.

Lastly, I want to thank my family and close friends for always believing in me and supporting me. Their love, guidance and motivation made a big difference to reach this milestone.

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In Brno, on December 24, 2024

Iskandar Mamasoliev



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1. Introduction

Understanding inequality is a critical issue within the European Union (EU). It influences economic stability, social cohesion, and opportunities for citizens. Since the region is characterized by diverse economic and political histories, measuring inequality across the EU presents unique challenges. This is because inequality can be examined across multiple economic dimensions that cannot be fully captured by any single metric. Moreover, each index is used for the purpose of condensing the complexities of distributions into one number (Brazauskas et al., 2023). Thus, using a range of measures is essential to gaining a comprehensive view of inequality in any region.

A variety of inequality measures are commonly used in economic analysis, each providing distinct insights into how income or wealth is distributed within a population. For example, the most frequently used one is Gini coefficient offers a summary measure of income distribution but is not easily decomposable at the depth of disparities within various income segments. In contrast, measures such as the Atkinson index and Generalized Entropy (GE) index incorporate parameters that adjust their sensitivity to specific parts of the distribution (Martín-Legendre, 2018). Each of these metrics offers unique strengths and limitations when together applied to a country or region's socioeconomic context.

This thesis applies these three inequality measures – Gini, Atkinson and Generalized Entropy – to analyze income inequality across all current 27 member states between 2007 and 2023 years. The EU as a political and economic union provides a valuable context for studying inequality due to its economic diversity. It includes both long-standing strong industrialized economies and newer member states that underwent significant transitions from socialist systems to market-oriented economies. Despite the continuous efforts to promote socioeconomic convergence since the EU's enlargement, income disparities still persist among some member states (Szczepaniak and Szulc-Obłóza, 2020).

By comparing trends across multiple measures, this study aims to identify patterns of inequality within and between countries. It highlights where inequality has worsened, improved or remained stable. Through this approach, the thesis tries to provide an objective perspective on income inequality and demonstrates the importance of using complementary measures to capture the full scope of disparities in a complex and evolving regional context.

1.1 Research Objectives and the Scope of Study

The primary goal of this study is to compare income inequality in the selected countries using various established measures. By applying multiple metrics, the research also aims to explore how these measures catch variations in income distribution and highlight regional differences within and across countries over time.

The key objectives of this research are as follows:

- To perform a comparative analysis of inequality and evaluate overall trends across the EU member states from 2007 to 2023 using three measures.
- To analyze the driver of changes by observing shifts in income shares across deciles where they widened or narrowed.
- To compare the results across the three measures and identify patterns of similarities or differences.

This research will begin with a literature review which covers some existing studies on the concept of “inequality” and the importance of this area from empirical research point of view. This is then followed by exploration of the selected measures, their theoretical formulations and characteristics. Based on these findings, the thesis will apply these measures to the most recent data from Eurostat and other relevant sources. It should be noted that this study will focus exclusively on income inequality rather than on other forms of inequality such as gender or educational disparities.

Through this structured approach the paper aims to provide a nuanced understanding of inequality trends in the EU. It explains diverse metrics to analyze regional disparities and assesses the effectiveness of those measures to understand the socio-economic imbalance within the Union.

2. Literature Review

This chapter begins with understanding income inequality and its measurements. The review is divided into three sections. The first section explains the different aspects of income inequality including its definitions, causes, and societal impacts. The second section introduces common measures of inequality and describes their construction, strengths and limitations. Then the third section examines income inequality in the European Union based on existing studies.

2.1 Understanding the Dynamics of Inequality

Inequality has been a central focus of economic research for the past many decades. It reflects disparities in access to resources, opportunities and well-being. Among the various dimensions of inequality, income inequality holds particular importance because it captures the uneven distribution of earnings, wages and other monetary resources within a population. Unlike wealth inequality which focuses on accumulated assets such as real estate and savings, income inequality measures the flow of resources over a short period (Tilly, 1991). This distinction allows it to reflect societal trends shaped by labor markets, structural shifts, and government interventions.

Research shows that high inequality carries significant societal consequences. The disparity can constrict social mobility, reduce economic growth, and widen the gap among social groups, as it is often seen in access to education, healthcare, and employment opportunities (Tsounta et al., 2015). Moreover, Wilkinson and Pickett (2010) state that highly unequal societies often experience lower social trust, higher crime rates and poorer mental and physical health. In contrast, they find that countries with the low inequality such as Scandinavian nations are inclined to have high levels of health and social cohesion which reinforces a process of governmental support and economic stability. Taken all together, equality serves as a precondition for greater trust to establish a stable democracy where citizens are more likely to participate actively and support equal governance.

In fundamental level, income inequality represents a shift from an ideally equal distribution where each individual in a society receives an identical share of resources. However, some argue that not all income differences can be considered “inequality” in a normative sense. Cowell (2009) discusses that some income variations may arise naturally for two basic reasons: need and merit which do not inherently reflect economic inequity. For example, larger families or individuals with health challenges often require greater resources to meet basic needs compared to single, young individuals. Similarly, skilled professionals who make great contributions to

society may receive higher compensation as a reflection of their productivity. He further explains that income differences are also linked to life cycles. Typically, younger individuals earn less than their older, more experienced counterparts who have accumulated skills and capital. While these differences seem to be natural, they often hide structural inequalities caused by unequal access to quality education, labor market discrimination or limited economic opportunities.

Understanding the drivers of income inequality helps to explain its root causes. Between the countries globalization and technological advancements have played dual roles in shaping inequality. While globalization has facilitated trade and economic growth, its benefits have often favored higher-income groups and skilled labor (Naoaj, 2023). Technological advancements, particularly automation, have further aggravated this trend by displacing low-skilled workers and concentrating income gains among capital owners. As automation reduced demand for routine jobs in the last century, similar fears now arise about artificial intelligence potentially displacing middle-skilled occupations in developed economies (Acemoglu, 2022).

The aspect of poverty further emphasizes the complexity of inequality. The term refers to a lack of sufficient resources to meet basic needs such as food, housing, and healthcare. It is often measured by the threshold of international poverty line or median income and is defined the share of the population living below these (Haughton & Khandker, 2009). Poverty serves as a visible outcome of income disparities, particularly in societies where inequality limits access to opportunities for lower-income groups.

It should be emphasized that this study focuses specifically on income inequality and its measures to analyze disparities within EU member states. While poverty and wealth inequality are related concepts, this research centers on income inequality due to its comparability across countries and the availability of methodological tools for measurement. A deeper understanding of income inequality is critical for informing policies that promote inclusive economic growth and address the persistent disparities within the European Union.

2.2 Common Inequality Measures

So, now we know that inequality is a social phenomenon and is essential to be studied, but what is indicated with its measure? Cowell (2009) states it as “a scalar numerical representation of the interpersonal differences in income within a given population”. Here with the word “scalar” he means all the aspects of inequality which are put into a statistical number or a single point. Suppose we have data of annual income (x_i where $i = 1, 2, \dots, n$) of n people. It will be easier to interpret that

incomes are sorted in ascending order among the population that is $x_1 \leq x_2 \leq \dots \leq x_n$. The question is to get a single measurement which defines each set of x_i in terms of inequality. At minimum it should be absolutely balanced or zero where all individuals have identical earnings and should have a positive indicator when the incomes of some differ (Allison, 1978). This condition is satisfied by almost all measures, but economists introduced several fundamental principles or axioms to get basic requirements for an appropriate inequality indicator.

Anonymity (Symmetry). An inequality measure Y is anonymous if for random 2 allocations, let's say $x, x' \in X$ and that x' is a permutation of x we obtain $Y(x) = Y(x')$. In other words, all individuals are interchangeable, and measure should not categorize them based on their identities, age or name. The value remains unchanged if the individuals in the distribution are permuted that the analysis focuses solely on their income (Neves Costa & Pérez-Duarte, 2019).

Population Invariance. Imagine we are handing out a disproportionate slice of pie at a party. This principle tells us that the pie distribution does not depend upon the number of pie receivers ($Y(x) = Y(\alpha x)$ for any allocation $x \in X$ and any real number $\alpha > 0$). Cowell (2009) states that if we assess inequality in population with n individuals and then combine it with another identical economy and maintain the same income distribution proportions, the result will be a total population of $2n$. If the inequality value remains consistent during such replication, then the measure meets the principle of population. However, he later argues that this axiom sometimes might be misleading. He gives an example of a two-person world (a) in which one person gets all the money and the other has none. Now creates an identical world and merges them together so that in an economy of four people (b): two are left poverty-stricken and the other two share income equally. He points out that (a) and (b) do not seem to be "equally unequal", despite getting the same measurement in both cases.

Principle of transfers (Pigou–Dalton). A measure of inequality Y fulfills Pigou–Dalton transfer axiom for any two allocations $x, x' \in X$ with distinct indices i, j , and $\alpha > 0$ considering $x_i \leq x_j$ and the transfer $x'_i = x_i + \alpha$ and $x'_j = x_j - \alpha$ while $x_n = x'_n$ for all $n \neq i, j$ it holds that $Y(x) > Y(x')$. This means the inequality measure Y decreases after the transfer. Specifically, a positive transfer from the rich to the poor in the distribution, where the rich still have more income than the poor, should lead to a reduction in inequality (Andonie et al., 2019).

Scale Invariance (Relative Income Principle). Let's return to our 'pie' example to illustrate this principle. And it says that generally the inequality of the slices should not rely on the pie's overall size. If we change all incomes by the same proportion, it can be then said that the income distribution remains fundamentally unaffected, and therefore the new inequality measure should be in its original value (Cowell, 2009).

These four properties ensure that the indicators behave in a logical and consistent manner. They are also shared by most common inequality measures, including those we will examine in the next chapter. However, it is worth noting that some indicators do not fully meet these principles, and this fact alone will not be sufficient to rule them out. Such measures are often used in specific contexts, particularly for subgroup or within-group analyses in a population, where they satisfy the decomposability principle (Neves Costa & Pérez-Duarte, 2019).

2.2.1 Selected measures

Several summary measures of inequality have been developed from the last century, and this section examines three widely used inequality measures. The interpretation of these approaches raises some distinct comparisons and questions about the meaning and interpretation of inequality.

First, we explore the Gini coefficient and its derivative from the Lorenz curve, which gives an intuitive and graphical representation of income distribution. Next, we consider the Atkinson index, and it offers a flexible way to measure inequality based on different degrees of aversion. Finally, we analyze Theil's both entropy measures (T and L indices).

Gini coefficient

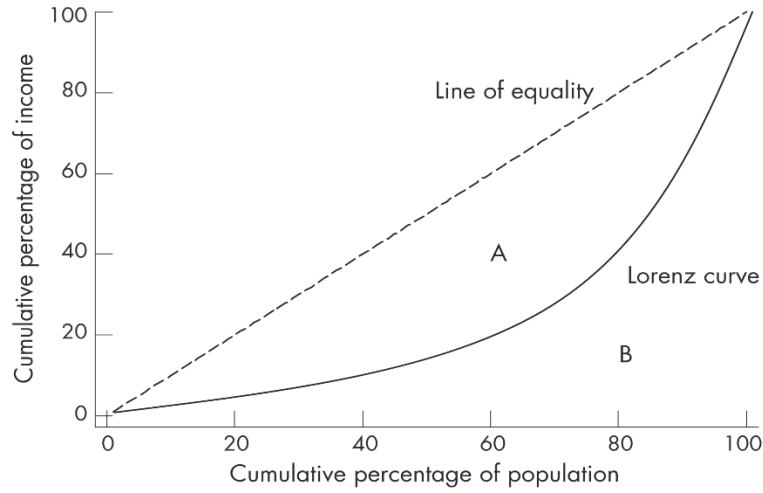
By far the most popular single inequality measure is the Gini coefficient. It was originally devised by an Italian mathematician named Corrado Gini in 1912. It is a single scalar value with a range between 0 and 1, where 0 represents perfect equality (everyone has the equal income) and 1 indicates absolute inequality (one individual possesses all the income or wealth). The Gini coefficient can also be expressed as a percentage, scaling from 0% to 100% (Sitthiyot & Holasut, 2020).

In practical applications, the Gini coefficient, G is given by:

$$G(Y_t) = \frac{1}{2n^2\mu_t} \sum_{i=1}^n \sum_{j=1}^n |y_{it} - y_{jt}| \quad (1)$$

where $Y_t = (y_{1t}, y_{2t}, \dots, y_{nt})$ and $\mu_t = \frac{1}{n} \sum_{i=1}^n y_i$

Here y_{it} is the income of an individual (i) at time t . Y_t is the income distribution in time period t and μ_t is the mean income of the distribution across individuals.

Figure 1. The Lorenz curve framework (hypothetical illustration).

Pascoal & Rocha (2017) point out that the Gini coefficient can also be derived from the Lorenz curve framework. The graph represents a cumulative distribution function and is defined mathematically as the cumulative share of total income (or wealth) attributed to cumulative shares of the population (see Figure_1). The dashed diagonal line shows a 45° line of equality, an ideal situation where each individual has the same income and therefore $G(Y_t) = 0$.

Graphically speaking, the Gini coefficient is calculated from the ratio of the area between the Lorenz curve and the 45° line of equality to the total area beneath that line. In Figure_1 these sections are named with “A” and “B”.

Thus,

$$G = \frac{A}{A + B} \quad (2)$$

If we assume the Lorenz curve is a finite discrete function, the area between the Lorenz curve and the line of equality can be approximated as a frequency polygon with the following formula (Esmail and McCloughan, 2003).

$$G = 1 - \sum_{i=1}^n (x_i - x_{i-1})(y_i - y_{i-1}) \quad (3)$$

Here, n is the number of intervals into which the population is grouped. x_i represents the cumulative share of income, whereas y_i is the cumulative share of the population.

Moreover, if the Lorenz curve is a monotonic increasing, twice differentiable, and convex function $f(x)$, where x denotes the cumulative share of income, the Gini coefficient can be calculated as follows (Martín-Legendre, 2018):

$$G = 1 - 2 \int_0^1 f(x) dx \quad (4)$$

Atkinson index

A British economist, Anthony Atkinson (1970) developed a class of inequality measures. His work allows changing sensitivity to inequality in different areas of the income distribution. It is achieved with a weighting parameter ε which measures the aversion level. The parameter reflects the societal value placed on income transfers between different groups. Higher values mean greater sensitivity to changes in the lower end of the income distribution. Unlike other measures, the Atkinson Index allows for a degree of subjectivity by enabling policymakers and researchers to specify the level of inequality (Saint-Jacques et al., 2020).

Given a population with values $(y_1, y_2, \dots, y_n)$ and an aversion parameter $\varepsilon > 0$, the Atkinson index is as follows:

$$A(\varepsilon) = 1 - \left(\frac{1}{n} \sum_{i=1}^n \left(\frac{y_i}{\bar{y}} \right)^{1-\varepsilon} \right)^{\frac{1}{1-\varepsilon}} \quad (5)$$

where:

- y_i is the individual income,
- $\bar{y}$ represents the mean income,
- ε is the parameter of inequality aversion, and $\varepsilon \neq 1$
- n is the total number of the population.

Since $\frac{1}{1-\varepsilon}$ is undefined at $\varepsilon = 1$, the Atkinson index can be expressed as:

$$A(1) = 1 - \frac{\prod_{i=1}^n \left(y_i^{\left(\frac{1}{n}\right)} \right)}{\bar{y}} \quad (6)$$

Similar to the Gini, The Atkinson Index ranges from 0 (perfect equality) to 1 (maximum inequality) and its value rises as income inequality becomes more severe or as ϵ value is increased.

To sum it up, the measure answers the following question: “How much fraction of income would society be willing to sacrifice to get an ideal egalitarian distribution?” And this percentage represents the Atkinson index (Gyan, 2022). For example, an Atkinson index of α indicates that people would be willing to sacrifice that much of their income to make the income distribution more even.

Generalized Entropy Measures

Another class of inequality indices that provide a flexible framework for analyzing income or wealth disparities within a population is The Generalized Entropy (GE) measures. De Maio (2007) notes that similar to Atkinson index, the flexibility of GE measures lies in the sensitivity parameter α that adjusts the emphasis placed on inequalities across different segments of the income distribution.

The general formula of the family of GE measures can be given by:

$$GE(\alpha) = \frac{1}{\alpha(\alpha - 1)} \left(\frac{1}{n} \sum_{i=1}^n \left(\frac{y_i}{\bar{y}} \right)^\alpha - 1 \right) \quad (7)$$

where:

- y_i is the disposable income of an individual,
- $\bar{y}$ represents the population mean income,
- n is the size of the population, and
- α is the sensitivity parameter.

The $GE(\alpha)$ index theoretically ranges from 0 to ∞ , where 0 represents perfect income equality, and higher values indicate greater income inequality. And the most common values of α used widely in academic literature and research are 0, 1, and 2 (Haughton & Khandker, 2009).

However, it is clear from Eq. (7) that $GE(\alpha)$ is undefined for values of $\alpha = 0$ and 1. Because of that two more special cases have been obtained using the de L'Hospital's rule (Muszyńska & Wędrowska, 2018).

when $\alpha = 0$:

$GE(0)$ is known as the mean logarithmic deviation (MLD), also commonly mentioned as Theil's L index. It is defined as follows:

$$GE(0) = \frac{1}{n} \sum_{i=1}^n \ln \left(\frac{\bar{y}}{y_i} \right) \quad (8)$$

when $\alpha = 1$:

Likewise, another special case of $GE(\alpha)$ is referred to as the Theil inequality index or Theil's T index, named after a Dutch econometrician, Henri Theil who devised it in 1967. Theil's measure is defined as the difference between the maximum possible entropy and the observed entropy within the income or wealth distribution.

Neves Costa & Pérez-Duarte (2019) show that the entropy of the distribution of income shares represented by the vector $z = (z_1, z_2, \dots, z_n)$, where $z_i = y_i / \sum_{i=1}^n y_i$ denotes the share of income earned by an individual i (with y_i being the income of a person i), is calculated as:

$$H(z) = \sum_{i=1}^n z_i \ln \left(\frac{1}{z_i} \right) \quad (9)$$

The maximum possible entropy means a state of complete equality, where all individuals or households receive the same amount of income. In this scenario, $z_i = 1/n$ for all i and the entropy is given by $H(z) = \ln n$. On the other hand, the lowest entropy corresponds to complete income inequality, where one individual or household receives all the income ($z_i = 1$) and everyone else has nothing at all ($z_j = 0$ for $j \neq i$). In this case, $H(z)$ approaches zero as $z \rightarrow 0$.

Theil's measure $GE(1)$ is then defined as the difference between the maximum possible entropy and the observed entropy (Neves Costa & Pérez-Duarte, 2019):

$$GE(1) = \sum_{i=1}^n z_i \ln(nz_i) = \frac{1}{n} \sum_{i=1}^n \frac{y_i}{\bar{y}} \ln \left(\frac{y_i}{\bar{y}} \right) \quad (10)$$

In cases of $\alpha \leq 0$, the measure becomes more sensitive to changes in the income distribution that impact the lower tail while a higher (positive) value of α makes it more sensitive to inequalities at the upper end (Cowell, 2006). Although the sensitivity parameter α can theoretically take any real value between $-\infty$ and ∞ , Bellù & Liberati (2006) note that in practice, α is typically chosen to be positive. This is because, when $\alpha < 0$, this class of indices becomes undefined if any incomes in the distribution are equal to zero.

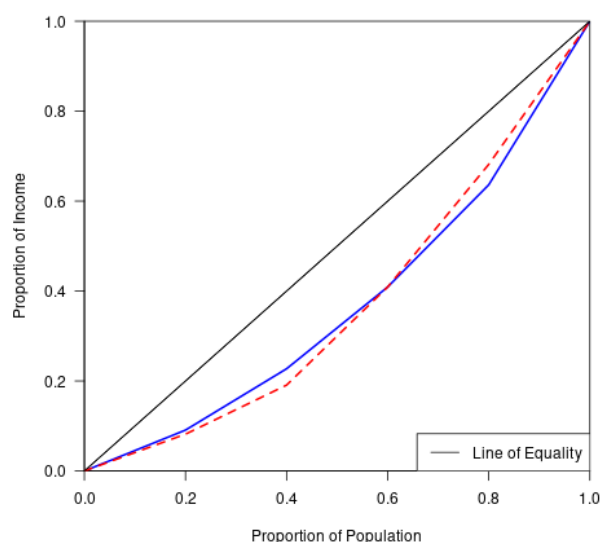
2.2.2 Strengths and limitations of the selected measures

Each inequality measure we have reviewed in the previous section – the Gini coefficient, the Atkinson Index, and the Generalized Entropy measures – offers unique advantages and faces inherent limitations. They all have distinct theoretical framework and methodological approaches in their construction. Therefore, the interpretation of their value varies from one to another even though they all measure inequality. While the Gini coefficient is celebrated for its intuitive interpretation, the Atkinson Index gives a normative dimension by incorporating societal preferences for inequality aversion, and the GE measures provide analytical flexibility and decomposability. This section further examines these measures to highlight their comparative versatility and downsides.

Among the three, the Gini coefficient is arguably the most employed inequality measure in official reports, and it is for a good reason. Sitthiyot & Holasut (2020) remark that straightforward interpretation and visualization with the Lorenz curve make it a very simple inequality measure among the international organizations, for instance, Eurostat, OECD and the World Bank run their own estimates for inequality index with Gini.

Despite its merits, Atkinson and Bourguignon (2015) argue that a lower Gini index in one country does not necessarily indicate a more equal distribution compared to another country with a slightly higher value. This is due to the possibility of intersecting Lorenz curves between these two countries. Cowell (2009) also explores this misleading concept, but in a case where two countries share the identical Gini value with distinctive income distributions (see Figure_2). He points out that it is due to loss of sensitivity of the measure to the information about different parts of the distribution.

Figure 2. Two Lorenz curves with the same Gini coefficient (hypothetical illustration).

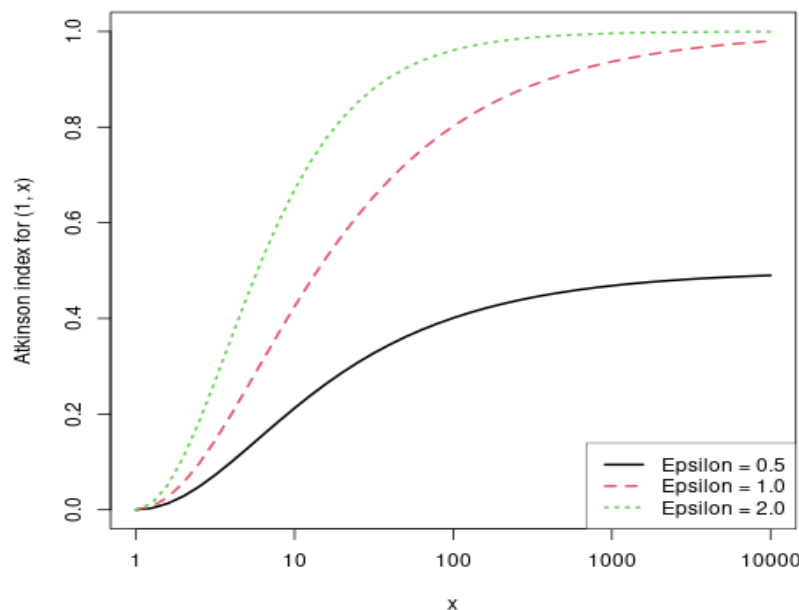


This limitation makes the statistical measure lack a direct welfare-based interpretation which can be seen in the A_ε index. This was particularly significant to Atkinson, who was concerned with the Gini framework's inability to give varying weights into the different parts of the income spectrum (De Maio, 2007).

Theoretically the aversion parameter in Atkinson index ranges from 0 to ∞ . Although the epsilon can be any non-negative number, the widely chosen parameters are 0.5, 1 and 2. For example, when $\varepsilon = 0.5$, the aversion becomes moderate to inequality and when $\varepsilon = 2.0$, it is much higher (Dutt & Tsetlin, 2020). So, as ε increases, society places greater emphasis on income transfers benefiting those at the lower end of the income distribution, and less weight to income transfers at the upper end (Martín-Legendre, 2018). This notion is clearly seen in Figure_3 where we are plotting this behaviour of several ε parameters for hypothetical data of $\{1, x\}$. Thus, the larger the ε , the greater the aversion to inequality.

According to Atkinson (1970), the Gini index generally provides rankings that are similar to those obtained with a relatively low level of inequality aversion.

Figure 3. Atkinson index for various parameters.



Similarly, the sensitivity parameter α of Generalized Entropy Measures enables to emphasize specific segments of the income distribution. The measure is mostly applauded for its decomposability feature. It is a within-group and between-group analysis of subgroup studies that is used for inequality comparisons of different regions, age or genders (Martín-Legendre, 2018). With that being said, Bellù & Liberati

(2006) comment that the meaning of $GE(\alpha)$ value may not be intuitive compared to Gini, primarily due to the varying behaviour of its parameter α . This may pose some challenges in communicating results to policymakers or the general audience.

Neves Costa & Pérez-Duarte (2019) further explore the topic by discussing the relationship between the Atkinson index (A_ε) and the GE indices. They explain that the Atkinson index can be converted into Generalized Entropy (GE) indices through a strictly increasing transformation, expressed as:

$$GE(1 - \varepsilon) = \frac{(1 - A_\varepsilon)^{1-\varepsilon} - 1}{\varepsilon(\varepsilon - 1)} \quad (11)$$

This relationship indicates that the Atkinson index with $\varepsilon > 0$ and the corresponding $GE(\alpha)$ index with parameter $\alpha = 1 - \varepsilon < 1$ generate the same ranking of income distributions. However, they also add that this transformation does not preserve the decomposability property of the GE indices.

Sitthiyot & Holasut (2020) report that there are currently over 50 indices to measure socioeconomic inequality in income and wealth distribution. From that, we have reviewed three well-known relative measures of inequality in the previous chapters along with their formulation of objectives and properties. In the next chapters, we develop a methodology for their application and attempt to interpret the resulting findings.

2.3 Inequality in the Context of the EU

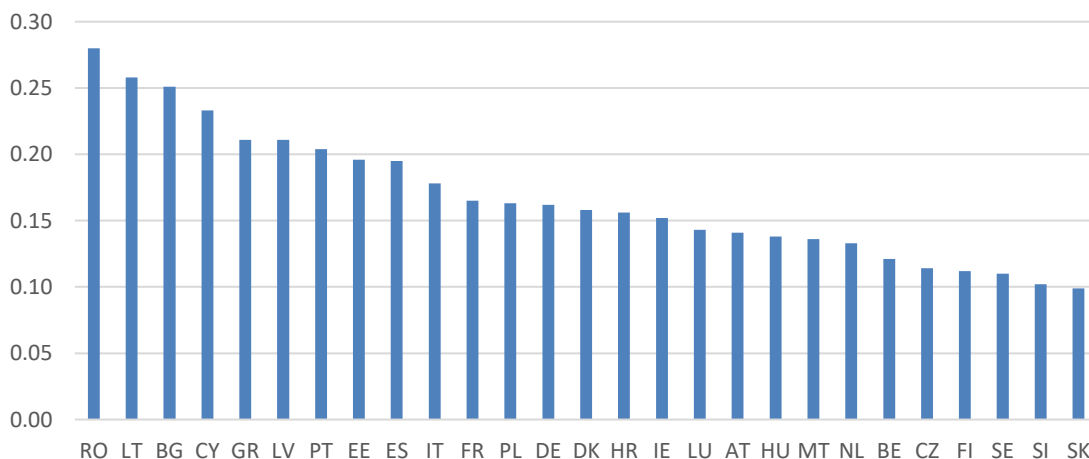
Income inequality in the EU has followed dynamic and often divergent trends caused by economic problems, policy responses and enlargement. Brandolini (2007) showed how the new Eastern European nations had increased overall economic disparity in the Union, but he also concluded that inequality was still lower than it was in the US. While the pre-2008 period showed progress toward convergence, researches show that inequality patterns shifted after the recession.

Before the financial crisis income inequality in the EU declined steadily. Research by Troitiño Cobas (2008) found a generally negative relationship between economic development and inequality in the EU-15 (pre-2004 enlargement). She concluded that countries with higher average incomes such as those in Northern Europe experienced lower inequality due to their redistributive policies. Likewise, in Eastern European countries such as Poland, Hungary and the Baltic states, Filauro (2018) also stated rapid economic growth and attributed this economic progress to rising wages

and improved labor market integration. However, after 2008 the economic downturn disrupted this process and exacerbated disparities in certain regions. Mediterranean economies such as Greece and Spain saw significant declines in real income levels particularly among low income households, while greater economies like Germany and France remained relatively resilient (Eurofound, 2017).

The Central and Eastern European (CEE) countries present an interesting contrast. Brzezinski (2019) noted rising inequality during 2008 and 2012 particularly in countries Hungary, Estonia, and Bulgaria where the Gini coefficient for disposable income increased due to falling employment levels. Moreover, Muszyńska and Wędrowska (2018) applied the Generalized Entropy measures to the Visegrád Group (V4) countries and highlighted significant reduction in inequality for Poland. They noted that it is driven by economic growth and rising wages that benefited middle-income groups in the country. However, their findings also showed Hungary continued to show higher GE values compared to other V4 members.

Figure 4. Theil index in the EU by country in 2014.



Source: Kranzinger (2019). The UK data is excluded.

Kranzinger (2019) observed significant disparities across the EU using the Theil index. In his findings (see Figure_4) Slovakia showed relatively equitable income distribution and in 2014 recorded the lowest Theil index value at 0.099 while Romania revealed underlining big disparities and recorded the highest value at 0.280. Other countries in his report with high inequality include Bulgaria and Lithuania whereas nations like Finland, Slovenia and the Czech Republic maintained much lower inequality. The four largest EU economies (Germany, France, Italy and Spain) together accounted for over 45% of the EU's total inequality which reflected their economic weight in the Union but also their internal disparities.

National redistributive policies play a central role to moderate inequality outcomes. Filauro (2018) noted that Western European nations known for their progressive taxation and robust social transfers, managed to maintain relatively low inequality levels in the last decade. In the meantime, despite improvements Southern European nations struggled with persistent disparities due to prolonged labor market disruptions. Similarly, Soukup (2023) finds this fluctuation in inequality with economic cycles where it aligns with Piketty's hypothesis that recessions tend to reduce inequality temporarily but begins to increase during recovery periods.

Structural drivers such as education and labor market access further explain regional differences. Muszyńska and Wędrowska (2018) emphasize the importance of education in reducing income disparities. The authors comment the Czech Republic and Slovakia's low inequality is due to their higher levels of educational attainment which enables access to well-paying jobs. Conversely, countries with lower levels of educational attainment experience noticeable disparities. For example, Bulgaria and Romania sustain some of the highest levels of inequality in the EU where low-skilled workers face stagnant wage structures.

In summary, the literature shows that income inequality in the EU varies considerably among member states and it is shaped by a mix of economic shocks, labor market structures and redistributive policies. While Nordic and Central European nations maintain relatively low inequality, rising income concentration among top earners remains a serious concern across the region. These findings underscore the importance of multi-dimensional analyses using measures like the Gini coefficient, Atkinson index and GE indices to capture the new inequality trends within the EU for the last decade.

3. Methodology

This chapter outlines the approach used to study income inequality across all European Union members. The EU currently consists of 27 member countries, and including all of them ensures a comprehensive and inclusive assessment of income inequality trends. The study accounts for the geographical, economic, and sociopolitical diversity within the EU and it ranges from founding members with highly industrialized economies to newer member states with transitional economies.

In the next section it shares data sources, and the methods employed for analysis. Next, it discusses the limitations of these data and methods to ensure the results are interpreted within the appropriate context.

3.1 Data Collection and Sources

The raw data utilized in the analysis comes from Eurostat (2024), the dataset titled “Distribution of income by quantiles”. This dataset consists of 180 worksheets and provides detailed disposable income distribution for all member states. The structure is composed of quartiles, quintile and deciles for income distribution in terms of Euro, Purchasing Power Parity (PPP), national currency and their share of national equivalised income.

For this analysis, the study focuses on income groups in deciles (10 equally divided clusters). The decision to work with deciles, as compared to other concepts, such as quantiles, enhances the level of granularity in the analysis and as a result allows for a more precise evaluation of income inequality. Moreover, the dataset includes Euro, PPP, national currency indicators and their respective shares for each decile. I have opted for the share of national equivalised income for each decile to calculate the inequality per country. It should be noted that this approach aligns with a critical axiom for all three measures mentioned in the literature review, the Scale Invariance. This principle leaves out disparities among respective national currencies and still provides an underlying relative inequality rate and thus maintains comparability across countries.

The dataset is structured records from 1995 for countries that were already members at that time. However, for countries that joined the EU later, data availability starts with their accession year. To ensure consistency across countries, the study focuses on the uniform time period from 2007 to 2023. This range reflects a balanced timeframe and also includes data for all member states excluding Croatia

which became an EU member in 2013, and the Eurostat dataset begins for this country from 2010. To account for the missing data prior to 2010, supplementary source is selected from the Croatian Bureau of Statistics in the report titled “*Rezultati Anketе o potrošnji kućanstava*”, APK (2010). This ensures that Croatia's inclusion in the analysis is as complete as possible, despite the limitation of Eurostat data before 2010.

3.2 Methods of Data Analysis

After cleaning and organizing the disposable income share data per decile, calculations are performed using the “*ineq*” library in R. It is a statistical package in CRAN repository specifically designed for inequality measurements (Zeileis & Kleiber, 2014). The processed results are then stored in an Excel file where additional analysis, charts, and graphs are constructed for visualization.

For the Gini coefficient, the study analyzes trends in income inequality for a selection of countries between 2007 and 2023. The results focus on identifying:

- Increasing or decreasing trends in income inequality over time.
- The magnitude of differences in that time period.

Following the discussion, the difference of two Lorenz curves in those countries are examined for changes within specific income subgroups. This comparison provides insights into which parts of the income distribution contributed most to the observed changes in overall inequality.

For the Atkinson Index and Generalized Entropy indices, the study assesses to what extent changes in their respective parameters influence the magnitude and interpretation of inequality:

- Atkinson Index: Different values of the inequality aversion parameter (ϵ) are applied ($\epsilon = 0.5, 1$ and 2). Note that higher values emphasize lower income groups and highlights how social aversion to inequality affects the measure's results.
- Generalized Entropy Indices: Sensitivity to the income distribution is evaluated using three commonly used parameters $\alpha = 0$ (Mean Log Deviation), $\alpha = 1$ (Theil Index), and finally $\alpha = 2$. These values capture income inequality in the lower, middle, and upper ends of the income distribution, respectively.

First, each measure and the relationship between their different parameters are discussed separately. Next, the similarities and parallels between the Atkinson and GE indices are analyzed to provide a comprehensive understanding of inequality measures. For calculation of both measures, the most recent data Eurostat (2024) is used.

The study then calculates the percentage change of all three inequality measures between 2007 and 2023 for selected countries. This comparative analysis explains the overall change in income inequality levels and the correlation among the selected measures. It also discusses the current state of average inequality in the EU.

Finally, bar charts and line graphs are used extensively to facilitate the interpretation of these results. After all, a well-made visualisation can tell stories in seconds what might otherwise take paragraphs (and a lot of brain nerves) to explain.

4. Data Analysis and Results

This section provides an examination of income inequality across the EU member states during the period 2007–2023. The analysis employs the Gini coefficient, the Lorenz curve, the Atkinson Index, and the Entropy indices to highlight trends, changes, and differences in inequality both over time and across these countries.

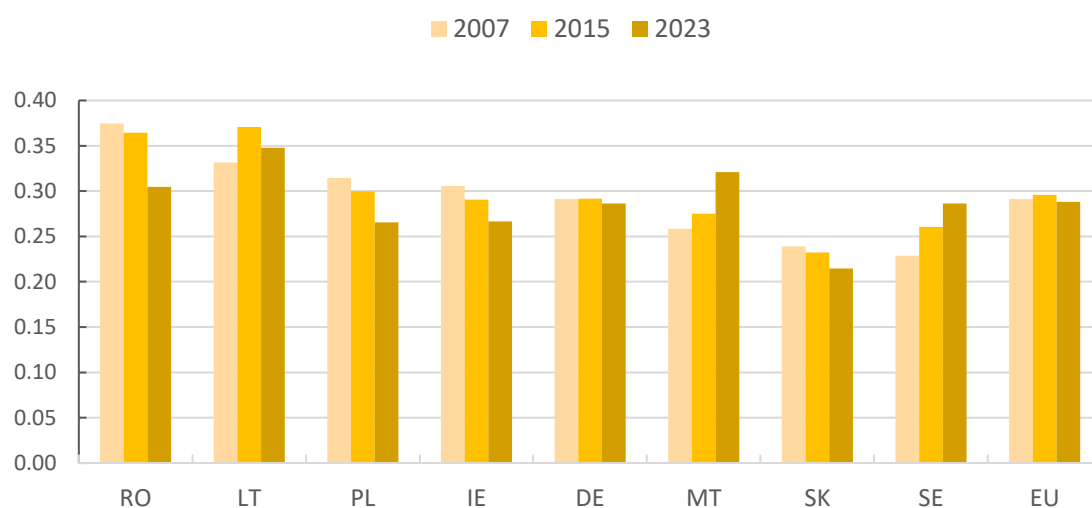
Through this analysis, the results aim to underline not only the overall trends in income inequality but also the significance of different income subgroups and the sensitivity of results to methodological choices.

4.1 Gini and the Lorenz Curve

The analysis begins with a display of inequality from 2007 to 2023 using the Gini coefficient. The measure, as introduced earlier, serves as a standard way of seeing inequality with insights into how income is distributed within a population. By observing both the changes and the countries' respective rankings in a particular year, it becomes possible to identify trends where some experience rising inequality, others demonstrate improvements, and a few show relative stability.

Figure_5 below reveals the Gini coefficients for a selection of EU countries. These countries were chosen to represent diverse trends which also reflect similar behaviour in other countries. The full findings are presented in Appendix I.

Figure 5. Gini coefficient for the selected countries from 2007 to 2023.



Note: Countries appear ranked according to the highest Gini value for 2007.

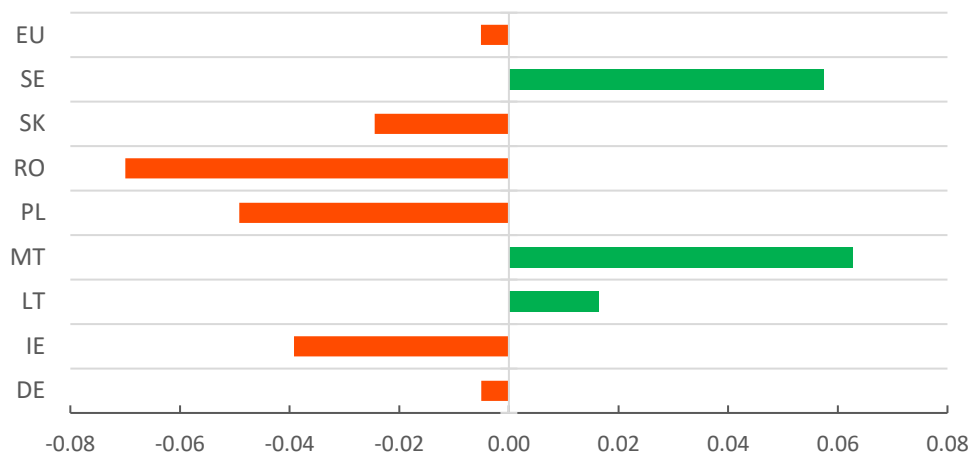
Source: own calculations

The data reveals several key insights. Romania, for example, saw the largest reduction in inequality. Its Gini coefficient decreased sharply from 0.375 in 2007 when it had the highest level of inequality among all EU member states (ranked 27th in 2007), to 0.305 in 2023 and improved its position to 18th. This significant progress highlights a narrowing income gap between low and high-income groups which we will talk about later in this section.

Poland also recorded a notable improvement over the last decade. The Gini coefficient fell from 0.315 in 2007 to 0.265 in 2023. As a result, Poland moved to the 7th place from 19th in 2007 and positioned itself among the most equal countries in the EU. This decline indicates a more balanced income structure, likely driven by economic reforms or redistributive policies. Countries such as Ireland and Slovakia followed similar positive trends. In Ireland, the Gini coefficient decreased from 0.306 in 2007 to 0.267 in 2023 and secured the 8th place. Slovakia began the period with one of the lowest levels of inequality with a Gini value of 0.239 in 2007. By 2023, the index had declined even further to 0.215 and Slovakia emerged the most equal country in the EU in terms of Gini. It should be noted that both the Czech Republic and Slovakia had had one of the lowest indices prior to EU accession and throughout the years in the Union, they managed to maintain a balanced income distribution or in some cases enhanced their rankings.

On the other hand, some countries experienced increasing inequality. Sweden, for example, was one of the most equal countries and in 2007 alone it was ranked 2nd. Its Gini was 0.229 in 2007 and by 2023 it increased to 0.286, lowering Sweden's position to the 12th place. This increase probably refers to widening income gap particularly in higher-income groups. A similar pattern occurred in Malta, where the index rose from 0.258 in 2007 to 0.321 in 2023. Malta's ranking dropped from 10th to 23rd during this period and placed it among the countries with higher inequality. In Lithuania, inequality increased more swiftly. The Gini grew from 0.332 in 2007 to 0.371 in 2015. In the next 8 years, the country was able to reduce the value to 0.348. Despite the signs of improvement, currently Lithuania is still standing close to the bottom of the EU ranking.

Although Figure_5 focuses on a subset of countries, the full dataset reveals additional trends across the EU. Countries such as Bulgaria and Latvia continue to show higher levels of inequality. In contrast, Finland and Spain maintained stable Gini values. For countries like Germany and France, income inequality remained relatively stable. Germany's Gini coefficient shifted only slightly from 0.291 in 2007 to 0.286 in 2023. Its ranking stayed consistent at the top half. Italy followed a similar pattern, with Gini value of 0.308 in 2023, showing no significant deviation from earlier levels.

Figure 6. Absolute change in Gini between 2023 and 2007.

Source: own calculations

Following the topic, the absolute changes in Gini coefficients between 2023 and 2007 as illustrated in Figure_6 provide further context to the trends observed earlier. Countries like Romania and Poland achieved the most significant reductions and their Gini values declined by approximately -0.07 and -0.05 respectively. These improvements align with their rising positions in the equality rankings too. Likewise, Ireland and Slovakia recorded smaller but meaningful reductions, reinforcing their status among the most equal EU countries.

In contrast, Sweden and Malta experienced the largest increases among the member nations with absolute changes of around +0.06. Meanwhile, Lithuania shows a mixed trend with an initial increase then followed by partial improvement.

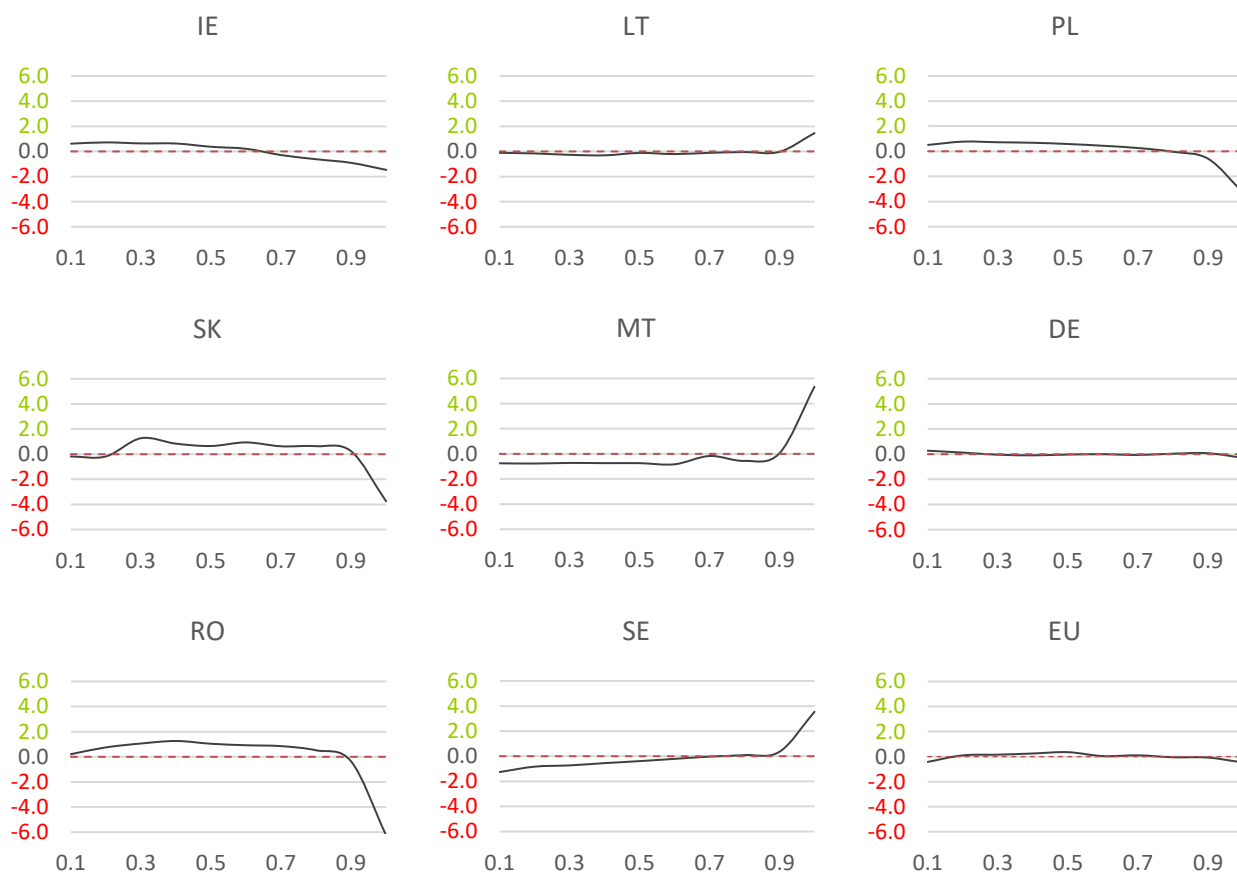
The EU average reflects minimal change and blindly suggests overall stability in income inequality across the region while masking regional improvements and setbacks at the national level which collectively shaped the overall EU trend.

These findings provide an overview of inequality trends across the EU. They identify countries that achieved progress in reducing inequality, countries that experienced growing income gaps, and those that remained relatively stable. However, to explain the drivers behind these changes, we need to examine shifts in the income shares distribution by comparing the difference of Lorenz curves for the selected countries.

Figure_7 below demonstrates the difference between the Lorenz curves for 2023 and 2007 across income deciles for the selected countries. The Lorenz curve as discussed earlier represents the cumulative income share held by different income

groups and provides a visual way to identify where shifts in income distribution occurred. The difference between the two curves reveals which deciles gained or lost income share in the distribution over the study period.

Figure 7. Difference between Lorenz curves from 2007 to 2023.



Source: own calculations

Here X-axis represents deciles in the income distribution and Y-axis shows deviation in income share between 2007 and 2023.

The results show distinct patterns among countries that experienced significant changes in the Gini coefficient. In Romania and Poland where we saw the Gini declined substantially, the lower income deciles (0.1–0.5 in X-axis) obtained income shares while the top decile saw notable losses. For example, in Romania, the income shares of the lowest decile increased by 0.2 percentage points, while the top lost approximately -6.3. Similarly, Poland's top decile saw a reduction of -3.3 points and the corresponding gains concentrated in the lower part of the distribution. These shifts indicate a redistribution of income from higher-income groups to the lower and highly support the observed reduction in the inequality.

In contrast, Sweden and Malta where inequality became worse the trends were reversed. Both countries saw significant gains in the top decile, alongside losses in the lower and middle deciles. For instance, Malta's top decile gained massive +5.3 percentage points while the bottom decile lost -0.8. Sweden followed a similar trajectory with the top decile gaining +3.6 percentage points and the lowest decile losing -1.3 percentage points. These patterns suggest that rising inequality in these countries was driven by a widening gap between higher and lower-income groups.

Other countries, such as Ireland and Slovakia, display more balanced changes across deciles. While the top income groups experienced moderate declines, the lower and middle deciles gained slightly, and this explains the gradual reduction in their Gini coefficients. For instance, in Slovakia, income share losses in the top decile were offset by gains in the lower deciles especially between 0.3 and 0.6.

For the EU, the differences between the Lorenz curves remain minimal across all deciles and it is also consistent with the earlier observation of overall stability in inequality with the Gini. However, this stability is merely the result of smoothed national variations observed in countries like Romania, Poland, Sweden, and Malta.

4.2 Atkinson and Generalized Entropy Indices

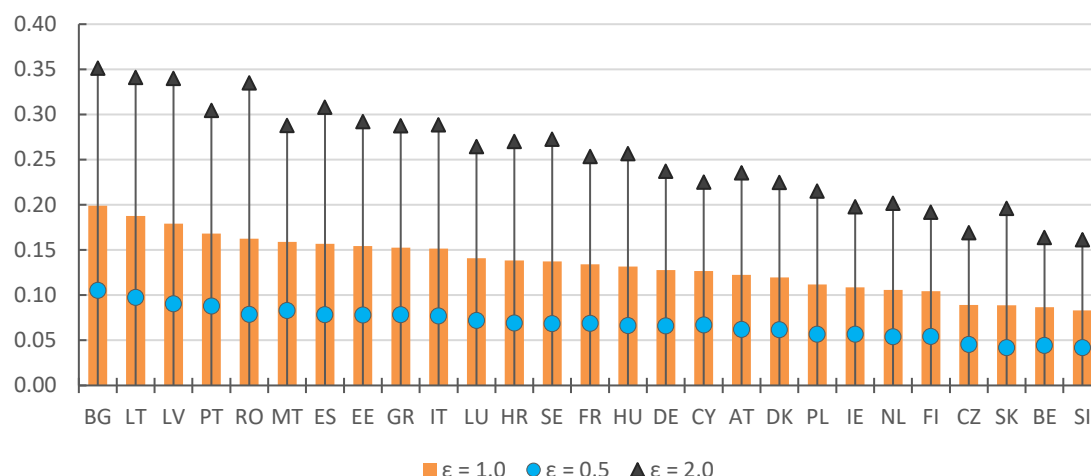
The analysis of income inequality continues with the Atkinson and the Generalized Entropy Indices. Two measures complement the Gini coefficient by offering sensitivity to different parts of the income distribution. While both indices quantify inequality, their results vary depending on specific parameters and weigh income disparities differently. These parameters allow the indices to emphasize income gaps at either the lower or upper ends of the distribution that reveals nuances a single measure may overlook.

Figure_8 below shows the Atkinson indices for the EU-27 countries in 2023 with the values reported for $\epsilon = 0.5$, $\epsilon = 1.0$ and $\epsilon = 2.0$. The chosen parameters respectively represent low, moderate and high sensitivity to inequality. It should be stated that as ϵ increases the index places more emphasis on income disparities at the lower end of the distribution.

The chart highlights clear trends. Countries like Bulgaria, Lithuania, and Portugal consistently rank among the most unequal across all sensitivity levels. For example, in Bulgaria, the Atkinson index increases sharply from 0.106 at $\epsilon = 0.5$ to 0.351 at $\epsilon = 2.0$. This steep rise reflects severe income disparities at the lower end of the distribution where lower-income groups hold disproportionately small shares of total

income. Similarly, Lithuania's Atkinson index rises from 0.098 to 0.341. The full findings can be found in Appendix I.

Figure 8. Atkinson indices for the EU-27 countries (2023).



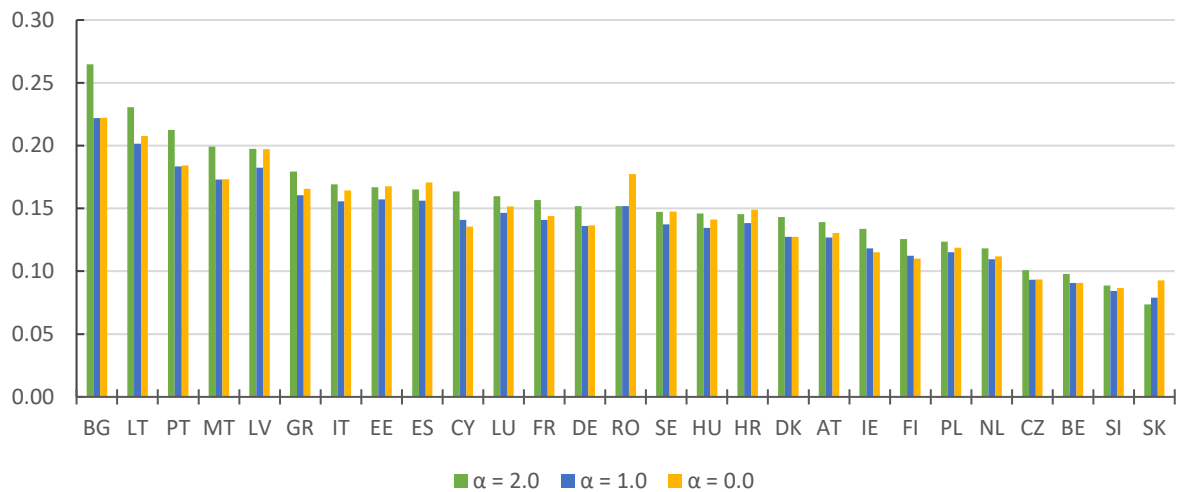
Note: Countries appear ranked according to the highest Atkinson index ($\epsilon=1$).

Source: own calculations

In contrast, countries such as Slovenia, Belgium and the Czech Republic exhibit relatively low Atkinson values, even as ϵ increases. Slovenia and Belgium maintain indices below 0.05 at $\epsilon = 0.5$, and their values remain moderate at 0.161 and 0.164 respectively for $\epsilon = 2.0$. These results suggest that income distribution in these countries remains balanced even when disparities at the lower end are weighted more heavily.

Interestingly, the Atkinson index also highlights subtle contrasts between countries with similar average inequality but differing sensitivity to lower income groups. For instance, Romania has higher values for $\epsilon = 2.0$ compared to Portugal (0.335 vs. 0.305). It suggests greater inequality in the lower-income deciles. However, for $\epsilon = 0.5$ and $\epsilon = 1.0$, Portugal consistently surpasses Romania. It implies that between these two countries Portugal's inequality is more evenly spread across the distribution while lower income groups in Romania are disproportionately disadvantaged.

The other measure in our analysis is the Entropy Index. It offers another perspective on income inequality by using the parameter α to adjust sensitivity. While $\alpha = 0.0$ treats all income disparities equally, $\alpha = 1.0$ (Theil index) weights income gaps proportionally. At $\alpha = 2.0$, the GE index becomes highly sensitive to income disparities at the upper end of the distribution and emphasizes the influence of higher income groups.

Figure 9. General Entropy indices for the EU-27 countries (2023).

Note: Countries appear ranked according to the highest GE index ($\alpha = 2.0$).

Source: own calculations

Figure_9 ranks the EU-27 countries by their GE values for $\alpha = 2.0$ providing insight how income inequality evolves as we shift the sensitivity. Once again, Bulgaria and Lithuania emerge as the most unequal countries at the top of the ranking. Bulgaria's GE value remains high across all parameters. It increases from 0.222 at $\alpha = 1.0$ to 0.265 at $\alpha = 2.0$. The rise highlights that its income disparities are largely concentrated at the upper end of the distribution where the richest continue to accumulate a significant share of total income. Similarly, Lithuania shows an upward trend with values rising from 0.208 to 0.231. The full findings are given in Appendix I.

In contrast, Romania presents an opposite behaviour. While Romania also reports high inequality, its GE index drops from 0.177 at $\alpha = 0.0$ to 0.152 at $\alpha = 1.0$, and it stabilizes at the same level for $\alpha = 2.0$. This trend suggests that inequality in Romania is more concentrated at the lower end of the distribution where smaller income disparities have greater influence when all income differences are weighted equally. Once sensitivity shifts to higher-income groups, we can see that the measure no longer increases dramatically. It tells that top-income concentration is not as severe as in Bulgaria.

Moving to the middle of the ranking, countries such as France, Germany and Luxembourg show moderate levels of inequality. For example, France's GE index rises from 0.144 at $\alpha = 0.0$ to 0.157 at $\alpha = 2.0$ while for Germany it ranges from 0.137 to 0.152. These small increases reflect relatively stable income structures where inequality is spread more evenly across the distribution and with no extreme concentration at

the top or bottom. Luxemburg follows a similar trend with values between 0.152 and 0.160.

At the lower end of the ranking, Slovakia and Czechia stand out as one of the most equal countries in the EU. Both report low GE values, but their patterns differ slightly. In Slovakia, the GE index decreases from 0.093 at $\alpha = 0.0$ to 0.074 at $\alpha = 2.0$. It shows that income disparities are minimal, even among higher-income groups. Czechia, on the other hand, maintains similarly low values, but shows a small increase at $\alpha = 2.0$ to 0.101, indicating a minor concentration of income at the top.

These observations emphasize the importance of the α parameter to uncover the drivers of inequality. In countries like Bulgaria and Lithuania, inequality gets worse as sensitivity to the top income increases. Conversely, in Romania, the drop in GE values between $\alpha = 0.0$ and $\alpha = 1.0$ reveals that income disparities are more severe at the lower end with impact on poorer households disproportionately.

By analyzing the Generalized Entropy index across multiple sensitivity levels, the results reveal that inequality in the EU is far from uniform. Some countries experience noticeable disparities at the lower end of the income distribution, while others face widening gaps at the top.

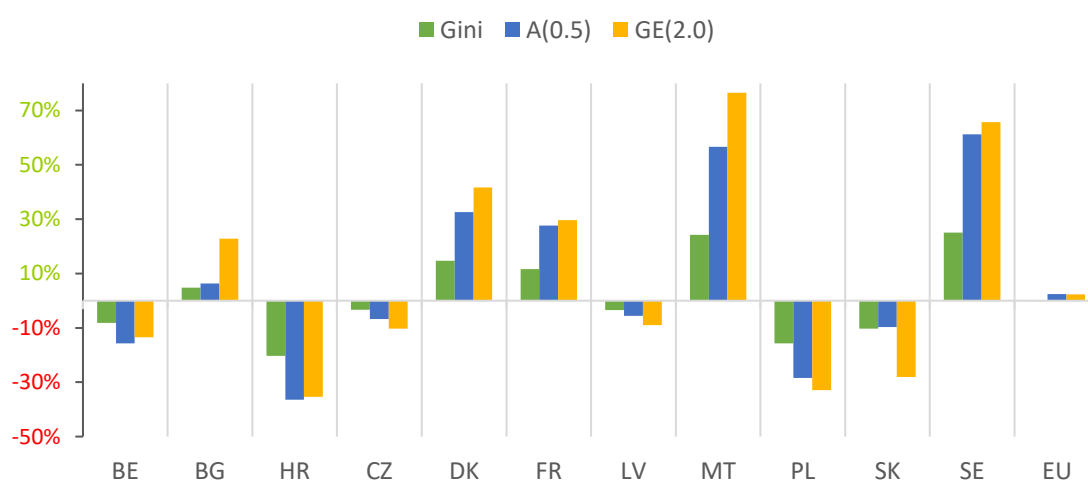
4.3 Relationship between the Measures

The relationship between the measures provides a more nuanced understanding of inequality dynamics. Although the Gini coefficient smooths over extremes and emphasizes inequality mainly in the middle of the distribution, it offers a balanced but generalized overview.

This generality often results in Gini giving results that fall somewhere between the Atkinson and GE indices. For example, countries like France and Germany where inequality is relatively balanced across the distribution, the Gini provides a similar trend to both the Atkinson at $\varepsilon = 1$ and the GE index at $\alpha = 1$. However, when inequality becomes concentrated at extreme, the Gini begins to diverge. As we have seen in countries like Sweden and Malta, top income groups gained disproportionately as a result the GE index at $\alpha = 2$ reveals sharp increases in inequality compared to the Gini. Similarly, in countries like Romania and Croatia the Atkinson index at $\varepsilon = 2$ highlights clear reductions that are less noticeable in the Gini coefficient.

The Atkinson and GE indices expose where inequality is concentrated. This layered perspective allows for more interpretations as seen in Romania and Portugal where inequality patterns differ between lower and upper ends of the distribution. However, despite this similarity, the two measures are not identical. The Atkinson index focuses on societal preferences for equality, meaning it considers inequality aversion and evaluates how much income society is willing to “sacrifice” to achieve perfect equality. By contrast, the GE measure takes a more mathematical approach and decomposes inequality by assigning weights to income disparities across the distribution.

Figure 10. Percentage change in selected measures between 2007 and 2023.



Source: own calculations

If we turn to the percentage change in selected measures between the years of 2007 and 2023 in Figure_10, the data further illustrates the distinct perspectives offered by each index. The complete findings are presented in Appendix I.

In Sweden inequality went up significantly across all measures. GE(2.0) moved up by 65.7%, compared to a 61.3% rise in A(0.5) and 25.1% in Gini. This disparity reveals that rising inequality in the country was driven primarily by income concentration among the top-income groups. This idea is also supported by the difference of two Lorenz curves we have examined earlier in this chapter. By contrast, in Croatia inequality declined sharply with A(0.5) and GE(2.0) showing larger reductions. The sharper decline in Atkinson suggests that inequality reductions in Croatia were driven primarily by improvements for poorer households where income gaps narrowed more significantly. While GE value also decreased, indicating some progress at the top end, the changes were smaller compared to those benefiting lower-income groups.

For middle-performing countries like France and Denmark inequality increased moderately between 2007 and 2023, but the trends differ slightly across measures. In France the Gini coefficient rose by 11.6% while the Atkinson index(0.5) and GE(2.0) increased by 27.7% and 29.7% respectively. This indicates that inequality deepened across both lower and upper-income groups with a slightly stronger concentration at the top end. Similarly, in Denmark the Gini grew by 14.7%, while Atkinson index climbed to 32.7% and GE to 41.7%. This indicates widening income gaps that affected low income households but primarily were driven by rising wealth among the top groups.

In countries such as Poland and Slovakia, it is clear to see that GE declined more sharply than A(0.5). It implies reductions among high income groups. For Poland, the Gini fell by -15.6%, but GE(2.0) recorded a steeper drop of -32.9% and it proves significant improvements in wealth concentration at the top end.

Overall, these trends show the importance of using multiple measures to understand inequality. While the Gini coefficient offers a general overview, the Atkinson index focuses on lower income groups, and the GE index highlights top end disparities. When analyzed together, the measures reveal where inequality changed across the entire income distribution.

5. Discussion

Throughout the previous sections, the study aimed to compare income inequality in the selected countries using various established measures from 2007 to 2023. By applying multiple metrics, the research also explored how these measures catch variations in income distribution and highlight regional differences within and across countries over time.

Generally, the results align with insights from the literature review. For instance, Brandolini (2007) claimed that the EU enlargement had increased overall inequality due to the inclusion of economically diverse member states. This notion is evident in the persistently high inequality levels for countries like Bulgaria and Lithuania where they consistently ranked among the highest in inequality across all measures. While enlargement provided opportunities for economic integration and growth, it also exposed structural disparities that remain challenging for the EU. These results also show the limitations of expansion in uniformly reducing income disparities and how difficult it is to bring an economically diverse region together.

Having said that, however, inequality reductions were notable in the region for countries Poland and Romania. Poland's improvements reflected in declining Gini over the study period. It also aligns with Muszyńska and Wędrowska (2018) who measured the inequality in the country with GE indices and documented how economic growth and rising wages benefited middle-income groups as a result contributed to reduced inequality. Although Kranzinger (2019) identified Romania as having the worst inequality in the EU in 2014 when measured with the Theil index, our results reveal significant recent improvements, particularly among lower-income groups after the country's membership into the EU. These shifts suggest the impact of redistributive efforts and economic reforms and also evident in the difference of Lorenz curves observed in the results.

The Czech Republic and Slovakia consistently showed low inequality in our results. These two countries managed to maintain some of the lowest Gini coefficients in the last 20 years. The findings are supported by Kranzinger (2019) who noted Slovakia's balanced income distribution and by Muszyńska and Wędrowska (2018) who attributed their success to strong education systems and efficient labor markets. On the other hand, low inequality does not necessarily equate to being the strongest economies. These countries remain economically stable, however they fall short of the major EU nations in terms of both market output and salary levels. While they prevented widening gaps in income groups seen elsewhere, future challenges such as technological disruptions and demographic shifts could pose new risks.

Attention should also be paid to the performance of the main EU economies. Troitiño Cobas (2008) observed that the EU-15 nations historically maintained lower inequality due to their stronger welfare systems and redistributive policies. Filastro (2018) further emphasized the role of sustained wage growth and improved labor market integration particularly in Germany and France. In our results, these two countries similarly displayed relatively stable values for all measures in the last decade which indicates minimal changes in income inequality over time. This stability aligns with Eurofound (2017) which discussed that these economies were more resilient during economic downturns and managed to avoid significant increases in income disparities. In Southwest Europe, Spain and Italy displayed inequality levels higher than the EU average, though they remained relatively stable over the study period. However, Portugal performed worse with measures indicating higher inequality and slower progress in addressing income differences.

Sweden's performance presents an unexpected result in this study. The country which has long been seen as a model of equity saw a significant rise in inequality as shown by all three metrics. The difference between Lorenz curves and GE indices emphasize that this increase was driven primarily due to decrease in income shares among the low income deciles. This unfortunately raises concerns mentioned by Wilkinson and Pickett (2010) earlier in the literature review about the consequences of unequal society. Currently, inequality in Sweden still remains below the EU average, but the rising disparity trend likely should pose some important questions about the sustainability of its welfare model and the resilience of traditionally egalitarian societies.

6. Conclusion

After analyzing with multiple measures, we observed that overall inequality in the EU has shown a mixed path. While the average inequality levels have remained relatively stable in the region, this hides significant variations among member states. Some countries, like Germany and France consistently align with or perform better than the EU average due to robust welfare systems. However, others including Bulgaria and Lithuania struggle to tackle their inequality issues. Over the study period, major reductions in inequality were observed in some regions, particularly in some Eastern European countries. Nevertheless, the stability of the EU average indicates that while some member states have improved, others have made less progress or even regressed in that time which emphasizes the need for more cohesive and effective EU-wide strategies.

This research has also illustrated the importance of employing multiple measures to fully capture the subtleties of inequality. Many studies rely solely on the Gini coefficient, overlooking the deeper insights offered by the Atkinson and Generalized Entropy indices. Although it offers a broad overview, the Atkinson and GE measures have provided deeper insights into disparities at different income levels which the Gini overlooked. Together, these tools have enabled a more complete picture of income distribution trends across the Union.

The findings further imply the literature review that addressing inequality calls for policies suited to each member nation's unique economic and social conditions. Strengthening redistributive mechanisms, investing in education and fostering inclusive labor markets are only a few examples of the multifaceted approach policymakers should consider. By pursuing these strategies, the EU can move closer to achieving its goals of social cohesion and sustainable development.

In conclusion, income inequality in the region is a complex issue influenced by a wide range of economic, social, and policy factors. With complementary use of the Gini coefficient, Atkinson index and GE indices, this study has aimed to highlight both common patterns and distinctive national paths. Finally, a comprehensive approach to inequality remains important as the European Union deals with economic integration and current social difficulties.

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8. Appendices

I. Key findings of the selected inequality measures for EU-27 countries.

Measure:	Gini						Atkinson Index (2023)			GE Indices (2023)			Percentage change ('07-'23)		
Country	2007	Pos.*	2015	Pos.	2023	Pos.	$\varepsilon = 0.5$	$\varepsilon = 1.0$	$\varepsilon = 2.0$	MLD ($\alpha = 0.0$)	Theil ($\alpha = 1.0$)	GE index ($\alpha = 2.0$)	Gini	A(0.5)	GE(2.0)
Austria	0.256	8	0.266	9	0.274	10	0.062	0.122	0.235	0.131	0.127	0.139	7.2%	18.4%	17.6%
Belgium	0.257	9	0.255	5	0.236	3	0.044	0.087	0.164	0.091	0.091	0.098	-8.1%	-15.7%	-13.4%
Bulgaria	0.345	23	0.361	25	0.361	27	0.106	0.199	0.351	0.222	0.222	0.265	4.8%	6.4%	22.9%
Croatia**	0.367	26	0.299	16	0.292	16	0.069	0.138	0.270	0.149	0.138	0.146	-20.3%	-36.5%	-35.4%
Cyprus	0.290	14	0.326	19	0.289	14	0.067	0.127	0.225	0.135	0.141	0.164	-0.3%	-1.0%	0.5%
Czech Republic	0.247	5	0.245	3	0.239	4	0.046	0.089	0.169	0.093	0.093	0.101	-3.4%	-6.7%	-10.3%
Denmark	0.238	3	0.264	8	0.274	9	0.062	0.120	0.225	0.127	0.127	0.143	14.7%	32.7%	41.7%
Estonia	0.326	20	0.339	23	0.313	22	0.078	0.154	0.292	0.168	0.157	0.167	-3.9%	-8.0%	-16.5%
Finland	0.255	7	0.246	4	0.260	6	0.054	0.104	0.192	0.110	0.112	0.126	2.0%	4.2%	5.7%
France	0.260	11	0.284	13	0.290	15	0.069	0.134	0.254	0.144	0.141	0.157	11.6%	27.7%	29.7%
Germany	0.291	15	0.292	15	0.286	13	0.066	0.128	0.237	0.137	0.136	0.152	-1.7%	-4.6%	-3.7%
Greece	0.335	22	0.334	21	0.309	21	0.078	0.153	0.288	0.166	0.160	0.179	-7.5%	-13.1%	-13.6%
Hungary	0.250	6	0.276	11	0.284	11	0.067	0.132	0.257	0.141	0.135	0.146	13.5%	32.5%	30.5%
Ireland	0.306	16	0.290	14	0.267	8	0.057	0.109	0.198	0.115	0.118	0.134	-12.8%	-23.0%	-22.1%
Italy	0.313	18	0.317	18	0.308	19	0.077	0.152	0.289	0.164	0.156	0.169	-1.5%	-3.0%	-3.7%
Latvia	0.346	24	0.346	24	0.334	25	0.091	0.179	0.340	0.197	0.182	0.198	-3.5%	-5.5%	-9.0%
Lithuania	0.332	21	0.371	27	0.348	26	0.098	0.188	0.341	0.208	0.201	0.231	4.9%	10.0%	13.7%
Luxembourg	0.268	13	0.278	12	0.300	17	0.072	0.141	0.265	0.152	0.146	0.160	11.7%	25.9%	23.6%
Malta	0.258	10	0.275	10	0.321	23	0.083	0.159	0.288	0.173	0.173	0.199	24.2%	56.6%	76.5%
Netherlands	0.265	12	0.259	6	0.258	5	0.054	0.106	0.202	0.112	0.110	0.118	-2.4%	-5.3%	-12.2%
Poland	0.315	19	0.300	17	0.265	7	0.057	0.112	0.215	0.119	0.115	0.124	-15.6%	-28.5%	-32.9%
Portugal	0.361	25	0.333	20	0.330	24	0.088	0.168	0.305	0.184	0.184	0.213	-8.5%	-15.2%	-18.7%
Romania	0.375	27	0.364	26	0.305	18	0.079	0.162	0.335	0.177	0.152	0.152	-18.7%	-31.3%	-42.5%
Slovakia	0.239	4	0.232	1	0.215	1	0.042	0.089	0.196	0.093	0.079	0.074	-10.2%	-9.7%	-28.1%
Slovenia	0.229	1	0.240	2	0.229	2	0.042	0.083	0.161	0.087	0.084	0.089	0.0%	0.1%	0.3%
Spain	0.312	17	0.338	22	0.309	20	0.078	0.157	0.308	0.171	0.156	0.165	-0.9%	-1.0%	-1.9%
Sweden	0.229	2	0.261	7	0.286	12	0.069	0.137	0.273	0.148	0.137	0.147	25.1%	61.3%	65.7%
European Union	0.291	-	0.296	-	0.288	-	0.069	0.134	0.255	0.145	0.140	0.154	-1.0%	2.5%	2.4%

* Ranking of a country in a particular year. Lower the ranking, lower the Gini.

** Data is completed from Croatian Bureau of Statistics.

II. ISO Codes for the EU-27

Code	Country
AT	Austria
BE	Belgium
BG	Bulgaria
HR	Croatia
CY	Cyprus
CZ	Czech Republic
DK	Denmark
EE	Estonia
FI	Finland
FR	France
DE	Germany
GR	Greece
HU	Hungary
IE	Ireland
IT	Italy
LV	Latvia
LT	Lithuania
LU	Luxembourg
MT	Malta
NL	Netherlands
PL	Poland
PT	Portugal
RO	Romania
SK	Slovakia
SI	Slovenia
ES	Spain
SE	Sweden
EU	European Union

III. Abbreviations

Abbreviations	
CEE	Central and Eastern European
FDI	Foreign Direct Investment
GE	Generalized Entropy
MLD	Mean Log Deviation
PPP	Purchasing Power Parity
V4	Visegrád Group
OECD	Organisation for Economic Co-operation and Development

IV. Correlation coefficients of several inequality measures for the EU-27 in 2023.

	Gini	A(0.5)	A(1.0)	A(2.0)	MLD	Theil	GE(2.0)
Gini	–	0.9933	0.9837	0.9351	0.9826	0.9931	0.9742
A(0.5)	0.9933	–	0.9951	0.9567	0.9953	0.9960	0.9730
A(1.0)	0.9837	0.9951	–	0.9808	0.9998	0.9822	0.9455
A(2.0)	0.9351	0.9567	0.9808	–	0.9796	0.9272	0.8660
MLD	0.9826	0.9953	0.9998	0.9796	–	0.9828	0.9470
Theil	0.9931	0.9960	0.9822	0.9272	0.9828	–	0.9897
GE(2.0)	0.9742	0.9730	0.9455	0.8660	0.9470	0.9897	–

V. Selected measures

Gini	$G(Y_t) = \frac{1}{2n^2\mu_t} \sum_{i=1}^n \sum_{j=1}^n y_{it} - y_{jt} $
Gini (Lorenz)	$G = 1 - 2 \int_0^1 f(x) dx$
Atkinson	$A(\varepsilon) = 1 - \left(\frac{1}{n} \sum_{i=1}^n \left(\frac{y_i}{\bar{y}} \right)^{1-\varepsilon} \right)^{\frac{1}{1-\varepsilon}}$
Atkinson (1.0)	$A(1) = 1 - \frac{\prod_{i=1}^n \left(y_i^{\left(\frac{1}{n} \right)} \right)}{\bar{y}}$
MLD	$GE(0) = \frac{1}{n} \sum_{i=1}^n \ln \left(\frac{\bar{y}}{y_i} \right)$
Theil	$GE(1) = \frac{1}{n} \sum_{i=1}^n \frac{y_i}{\bar{y}} \ln \left(\frac{y_i}{\bar{y}} \right)$
Generalized Entropy	$GE(\alpha) = \frac{1}{\alpha(\alpha-1)} \left(\frac{1}{n} \sum_{i=1}^n \left(\frac{y_i}{\bar{y}} \right)^\alpha - 1 \right)$